Event probabilities in the C160 moment representation

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The first C160 block already certifies all sixteen event probabilities. Its features are the thirty-two multilinear Bernstein monomials in the five physical parameters. The certificate follows directly from the two-qubit outcome amplitudes, so no semidefinite optimization is needed.

Write $z = (k, a_0, b_0, a_1, b_1)$ and define

$$d(t) = (1-t)^2 + t^2, \qquad D = (1+k^2) \prod_{t \in \{a_0, b_0, a_1, b_1\}} d(t).$$

The state is $(|00\rangle + k|11\rangle)/\sqrt{1+k^2}$. For the observable

$$O(t) = \frac{1-2t}{d(t)} Z + \frac{2t(1-t)}{d(t)} X,$$

unnormalized eigenvectors for outcomes $+1$ and -1 are $f_+(t) = (1-t, t)$ and $f_-(t) = (-t, 1-t)$. They are orthogonal and each has squared norm $d(t)$. Thus the numerator of the amplitude for outcomes α, β at settings x, y is

$$P_{\alpha\beta xy} = f_{\alpha,0}(a_x) f_{\beta,0}(b_y) + k f_{\alpha,1}(a_x) f_{\beta,1}(b_y).$$

Born's rule gives the following four-square identity. Here $e_0(t) = 1-t$, $e_1(t) = t$, $\bar{x} = 1-x$ and $\bar{y} = 1-y$:

$$D\, p(\alpha, \beta | x, y) = \sum_{r,s=0}^{1} \left[P_{\alpha\beta xy} e_r(a_{\bar{x}}) e_s(b_{\bar{y}}) \right]^2. \tag{1}$$

Every root is multilinear in all five variables. Consequently it has a unique expansion in

$$\phi_\varepsilon(z) = \prod_{i=0}^{4} z_i^{\varepsilon_i} (1-z_i)^{1-\varepsilon_i}, \qquad \varepsilon \in \{0,1\}^5.$$

The coefficient of ϕ_ε is simply the root's value at the cube vertex ε. This gives four integer vectors $r^{rs}_{\alpha\beta xy} \in \mathbb{Z}^{32}$ and the explicit Gram

$$G_{\alpha\beta xy} = \sum_{r,s=0}^{1} r^{rs}_{\alpha\beta xy} (r^{rs}_{\alpha\beta xy})^{\mathsf{T}} \succeq 0.$$

All sixteen Grams have rank four.

Proposition 1. *Every point of the normalized C160 moment projection defines nonnegative, normalized and no-signalling binary outcome probabilities.*

Proof. Let ℓ be its moment functional, with $\ell(D) = 1$, and let $M_0(\ell) = [\ell(\phi_i\phi_j)] \succeq 0$ be the first block. The physical coordinate maps define q by $q_\nu = \ell(D\Phi_\nu)$. Applying ℓ to (**??**) gives

$$\frac{1 + \alpha q_{A_x} + \beta q_{B_y} + \alpha\beta q_{E_{xy}}}{4} = \mathrm{tr}(G_{\alpha\beta xy} M_0(\ell)) \geq 0.$$

Summing over both outcomes gives one. Summing over either outcome gives $(1 + \alpha q_{A_x})/2$ or $(1 + \beta q_{B_y})/2$, independently of the other party's setting. □

The sum of the four event Grams represents D. It need not equal the strictly positive normalization Gram $Q_D = \left(\begin{smallmatrix} I_{16} & I_{16} \\ I_{16} & 2I_{16} \end{smallmatrix}\right)$ entry by entry: Gram representations of the same polynomial are not unique. The exact check compares their polynomial coefficients.

The file `C160_EVENT_GRAMS.json` contains all sixty-four integer root vectors. The accompanying `check_c160_event_grams.py` reconstructs them from the amplitudes and verifies all sixteen identities against the actual 243-row C160 coefficient map, including its off-diagonal factors. These event certificates establish probability positivity. Equality of the C160 projection with the quantum set still requires certificates for every quantum Bell inequality.